

Enhancement Two Narrative: Algorithms and Data Structure

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Artifact Description

The artifact I selected for the algorithms and data structure category is the same text-based adventure game that I enhanced in the previous milestone. I originally created this project early in the Computer Science program, and the main purpose of the game is for the player to move through rooms, collect gemstones, and escape the villain. For this milestone, I focused on improving the algorithmic side of the game by adding a shortest-path feature. This feature did not exist at all in the original version, so it gave me a good opportunity to add a real algorithm to the project and show more advanced skills.

Justification for Inclusion

I included this artifact in my ePortfolio again because the new enhancement shows clear growth in my understanding of algorithms and data structures. The biggest improvement is the addition of a Breadth-First Search (BFS) algorithm that finds the shortest path between two rooms. The rooms in the game naturally form a graph, with each room being a node and each direction being a connection. The BFS algorithm uses a queue, a visited set, and a map that stores how each room was reached. This allowed me to build the shortest path step by step and then show the result to the player.

I also added a new “hint” command so the player can see a list of room names. This supports the BFS feature by making it easier for the player to know where they can navigate. Adding BFS made me realize that the player did not have an easy way to know the exact names of rooms. The hint command fixes that problem and improves the overall experience.

Another important improvement is that the BFS algorithm avoids the villain’s room unless the player specifically asks for a path to that room. This required some careful logic inside the

algorithm, and it shows that I can adjust an algorithm to match the rules of the game. These additions show that I can apply algorithms in a practical way and make sure they fit the project's needs.

Course Outcome Connection

In Module One, I planned for this enhancement to support the course outcome related to algorithms and data structures. I believe I met this outcome with the addition of the BFS shortest-path feature. This enhancement required me to use a real algorithm, structure the graph of rooms correctly, and manage data in a way that supports the search process. I also used a queue and a visited set, which are standard data structures used in BFS. Because I was able to apply these ideas successfully, I feel that this enhancement matches my original plan, and I do not need to change my outcome coverage.

Reflection on the Enhancement Process

Getting to work on this enhancement helped me understand how important algorithms and data structures are for solving real problems. Before adding BFS, the game could only move one step at a time, and there was no way to guide the player. Once I added the BFS algorithm, I saw how much more organized the movement logic became. It also made the game feel more interactive and helpful.

One challenge I faced was making sure the BFS algorithm did not use the villain's room as a shortcut. I had to add special logic to skip that room unless the player was trying to reach it. Another challenge was realizing that players might not know the exact room names they could use with the new path command. This led me to add the hint command, which shows the full list

of room names. This experience taught me how one enhancement can uncover another need in the program, and how important it is to think about the user experience as well.

Overall, this enhancement helped me grow my understanding of algorithms, data organization, and problem solving. It also showed me how algorithmic thinking can make a project stronger and more helpful to the user. This is why I believe this artifact and enhancement belong in my ePortfolio.